

Elements of Statistical Learning Solutions

Daniel Mitsutani

2 Overview of Supervised Learning

Exercise 2.1. Suppose each of K -classes has an associated target t_k , which is a vector of all zeros, except a one in the k -th position. Show that classifying to the largest element of $\hat{y}$ amounts to choosing the closest target, $\min_k \|t_k - \hat{y}\|$, if the elements of $\hat{y}$ sum to one.

Solution. Evidently, the closest target corresponds to the k which minimizes $(t_k^1 - \hat{y}^1)^2 + \dots + (t_k^K - \hat{y}^K)^2$, where superscripts denote vector components. By definition of t_k , this sum can be written as:

$$\sum_{j \neq k} (\hat{y}^j)^2 + (1 - \hat{y}^k)^2.$$

The difference of the (square) distance to t_j and the (square) distance to $t_{j'}$ is thus given by the differing terms, namely:

$$\|t_j - \hat{y}\|^2 - \|t_{j'} - \hat{y}\|^2 = (1 - \hat{y}^j)^2 - (1 - \hat{y}^{j'})^2 + (\hat{y}^{j'})^2 - (\hat{y}^j)^2 = 2(\hat{y}^{j'} - \hat{y}^j).$$

Hence we have $\hat{y}^j \geq \hat{y}^{j'}$ if and only if the distance to t_j is less than or equal to the distance to $t_{j'}$, and classifying to the largest component of $\hat{y}$ is equivalent to the distance criterion.

Exercise 2.2. Show how to compute the Bayes decision boundary for the simulation example in Figure 2.5.

Solution. Denote the classes g_1 and g_2 . By Bayes' theorem, we have $\Pr(g_i | X = x) \propto \Pr(X = x | g_i) \Pr(g_i)$. The constant of proportionality does not depend on g_i , and the marginal probabilities are equal: $\Pr(g_1) = \Pr(g_2)$. The decision boundary is thus $\Pr(X = x | g_1) = \Pr(X = x | g_2)$.

With no knowledge of the chosen centers of the distributions, the decision boundary would be the line $x_1 = x_2$, since the centers are drawn from distributions centered at $(0, 1)$ and $(1, 0)$. This is not what is obtained in Figure 2.5, so we deduce that the decision boundary in Figure 2.5 must have been obtained conditioning on the centers m_k^i for the classes g_i .

With known centers, we have:

$$\Pr(X = x | g_i) \propto \frac{1}{10} \sum_{k=1}^{10} e^{-5\|x - m_k^i\|^2/2}.$$

Thus the decision boundary is given by numerically solving the equation:

$$\sum_{k=1}^{10} e^{-5\|x - m_k^1\|^2/2} = \sum_{k=1}^{10} e^{-5\|x - m_k^2\|^2/2}$$

Exercise 2.3. Derive equation (2.24).

Solution. A ball of radius r has volume $r^p \text{Vol } B(1)$, where $B(1)$ is the unit ball. Hence the probability that a given point lies inside it is r^p . The probability that a given point lies outside it is $1 - r^p$; the probability that all points lie outside it is $(1 - r^p)^N$. The median smallest distance $d(p, N)$ is the radius r such that the probability above is $1/2$. Solving for r gives

$$d(p, N) = \left(1 - \frac{1}{2^{1/N}}\right)^{1/p}$$

Exercise 2.4. The edge effect problem discussed on page 23 is not peculiar to uniform sampling from bounded domains. Consider inputs drawn from a spherical multinormal distribution $X \sim N(0, \mathbf{I}_p)$. The squared distance from any sample point to the origin has a χ_p^2 distribution with mean p . Consider a prediction point x_0 drawn from this distribution, and let $a = x_0/\|x_0\|$ be an associated unit vector. Let $z_i = a^T x_i$ be the projection of each of the training points on this direction.

Show that the z_i are distributed $N(0, 1)$ with expected squared distance from origin 1, while the target point has expected squared distance p from the origin.

Hence for $p = 1$, a randomly drawn test point is about 3.1 standard deviations from the origin, while all the training points are on average one standard deviation along direction a . So most prediction points see themselves as lying on the edge of the training set.

Solution. This can be seen two ways. First, by rotational symmetry of the distribution of X , we can take $a = (1, 0, \dots, 0)^T$, so that $z_i = x_i^1$ which is evidently $N(0, 1)$. Second, we can write $z_i = a_1 x_i^1 + \dots + a_p x_i^p$. Since the $a_j x_i^j$ are i.i.d. $N(0, a_j^2)$, we know their sum is $N(0, \sum_j a_j^2) = N(0, 1)$, since a is unit norm.

Exercise 2.5. (a) Derive equation (2.27). The last line makes use of (3.8) through a conditioning argument.

(b) Derive equation (2.28), making use of the cyclic property of the trace operator [$\text{trace}(AB) = \text{trace}(BA)$], and its linearity (which allows us to interchange the order of trace and expectation).

Solution.

(a) Recalling that $\hat{y}_0$ is independent of y_0 , and y_0 is independent of $\mathcal{T}$, we obtain expanding

the expression:

$$\mathbb{E}_{y_0|x_0} \mathbb{E}_{\mathcal{T}}(y_0 - \hat{y}_0)^2 = \mathbb{E}_{y_0|x_0}(y_0)^2 - 2 \mathbb{E}_{y_0|x_0}(y_0) \mathbb{E}_{\mathcal{T}}(\hat{y}_0) + \mathbb{E}_{\mathcal{T}}(\hat{y}_0)^2.$$

Adding and subtracting $(x_0^T \beta)^2$, and recalling that $\mathbb{E}_{y_0|x_0}(y_0) = x_0^T \beta$, we see that this equals:

$$\text{Var}(y_0|x_0) + (x_0^T \beta)^2 - 2(x_0^T \beta) \mathbb{E}_{\mathcal{T}}(\hat{y}_0) + \mathbb{E}_{\mathcal{T}}(\hat{y}_0)^2.$$

Now we add and subtract $[\mathbb{E}_{\mathcal{T}}(\hat{y}_0)]^2$, and grouping the positive added term with $(x_0^T \beta)^2 - 2(x_0^T \beta) \mathbb{E}_{\mathcal{T}}(\hat{y}_0)$ yields:

$$\text{Var}(y_0|x_0) + [\mathbb{E}_{\mathcal{T}}\hat{y}_0 - x_0^T \beta]^2 + \mathbb{E}_{\mathcal{T}}(\hat{y}_0)^2 - [\mathbb{E}_{\mathcal{T}}(\hat{y}_0)]^2 = \text{Var}(y_0|x_0) + \text{Bias}^2(\hat{y}_0) + \text{Var}_{\mathcal{T}}(\hat{y}_0).$$

To justify the last line of (2.27), we only need to explain the variance term $\text{Var}_{\mathcal{T}}(\hat{y}_0)$. Recall that by definition $\hat{y}_0 = x_0^T \hat{\beta}$ and the variance of $\hat{\beta}$ given $\mathbf{X}$ is given by $\text{Var}_{\mathbf{y}|\mathbf{X}}(\hat{\beta}) = (\mathbf{X}^T \mathbf{X})^{-1} \sigma^2$ by (3.8). Hence $\text{Var}_{\mathbf{y}|\mathbf{X}}(x_0^T \hat{\beta}) = x_0^T (\mathbf{X}^T \mathbf{X})^{-1} x_0 \sigma^2$. Therefore:

$$\text{Var}_{\mathcal{T}}(\hat{y}_0) = \mathbb{E}_{\mathcal{T}} \text{Var}_{\mathbf{y}|\mathbf{X}}(x_0^T \hat{\beta}) = \mathbb{E}_{\mathcal{T}} x_0^T (\mathbf{X}^T \mathbf{X})^{-1} x_0 \sigma^2,$$

as desired.

- (b) We use commutativity of trace for matrices of different dimensions; namely if A is $(m \times n)$ and B is $(n \times m)$, then $\text{trace}(AB) = \text{trace}(BA)$ is still true. Applying this to $x_0^T (1 \times p)$ and $\text{Cov}(X)^{-1} x_0 (p \times 1)$, and the fact that the trace of a scalar is just the scalar, we have:

$$\mathbb{E}_{x_0} x_0^T (\text{Cov}(X))^{-1} x_0 = \text{trace}[\mathbb{E}_{x_0} (\text{Cov}(X))^{-1} x_0 x_0^T] = \text{trace}[\text{Cov}(X)^{-1} \text{Cov}(x_0)].$$

Assuming x_0 has the same distribution as the training data X , $\text{Cov}(X)^{-1} \text{Cov}(x_0)$ is the $p \times p$ identity, hence the trace is p .

Exercise 2.6. Consider a regression problem with inputs x_i and outputs y_i , and a parameterized model $f_{\theta}(x)$ to be fit by least squares. Show that if there are observations with tied or identical values of x_i , then the fit can be obtained from a reduced weighted least squares problem.

Solution. Suppose there are N data points of the form (x, y_i) . The square error of the model on these points is $\sum_i (y_i - f_{\theta}(x))^2$.

Exercise 2.7. Suppose we have a sample of N pairs x_i, y_i drawn i.i.d. from the distribution characterized as follows:

$$\begin{aligned} x_i &\sim h(x), \text{ the design density} \\ y_i &= f(x_i) + \varepsilon_i, \text{ } f \text{ is the regression function} \\ \varepsilon_i &\sim (0, \sigma^2) \text{ (mean zero, variance } \sigma^2) \end{aligned}$$

We construct an estimator for f linear in the y_i ,

$$\hat{f}(x_0) = \sum_{i=1}^N \ell_i(x_0; \mathcal{X}) y_i$$

where the weights $\ell_i(x_0, \mathcal{X})$ do not depend on the y_i , but do depend on the entire training sequence of x_i , denoted here by $\mathcal{X}$.

(a) Show that linear regression and k -nearest-neighbor regression are members of this class of estimators. Describe explicitly the weights $\ell_i(x_0, \mathcal{X})$ in each of these cases.

(b) Decompose the conditional mean-squared error

$$E_{\mathcal{Y}|\mathcal{X}}[(f(x_0) - \hat{f}(x_0))^2]$$

into a conditional squared bias and a conditional variance component. Like $\mathcal{X}$, $\mathcal{Y}$ represents the entire training sequence of y_i .

(c) Decompose the (unconditional) mean-squared error

$$E_{\mathcal{Y}, \mathcal{X}}[(f(x_0) - \hat{f}(x_0))^2]$$

into a squared bias and a variance component.

(d) Establish a relationship between the squared biases and variances in the above two cases.

Solution.

(a) For linear regression, $\hat{f}(x_0) = x_0^T \beta$, where $\beta = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T y$, so that

$$\ell_i(x_0; \mathcal{X}) = x_0^T (\mathbf{X}^T \mathbf{X})^{-1} x_i$$

For k -nearest-neighbors, we may write:

$$\hat{f}(x_0) = \frac{1}{k} \sum_{x_i \in N_k(x_0)} y_i,$$

so that:

$$\ell_i(x_0, \mathcal{X}) = \frac{1}{k} I_{N_k(x_0)}(x_i),$$

and $I_{N_k(x_0)}$ is the indicator function of the k -nearest neighbors of x_0 .

(b)

$$\begin{aligned} \mathbb{E}_{\mathcal{Y}|\mathcal{X}}[(f(x_0) - \hat{f}(x_0))^2] &= \mathbb{E}_{\mathcal{Y}|\mathcal{X}}[\hat{f}(x_0)^2] - 2f(x_0)\mathbb{E}_{\mathcal{Y}|\mathcal{X}}[\hat{f}(x_0)] + f(x_0)^2 \\ &= \mathbb{E}_{\mathcal{Y}|\mathcal{X}}[\hat{f}(x_0)^2] - \mathbb{E}_{\mathcal{Y}|\mathcal{X}}[\hat{f}(x_0)]^2 \\ &\quad + \mathbb{E}_{\mathcal{Y}|\mathcal{X}}[\hat{f}(x_0)]^2 - 2f(x_0)\mathbb{E}_{\mathcal{Y}|\mathcal{X}}[\hat{f}(x_0)] + f(x_0)^2 \\ &= \text{Var}_{\mathcal{Y}|\mathcal{X}}(\hat{f}(x_0)) + (\mathbb{E}_{\mathcal{Y}|\mathcal{X}}[\hat{f}(x_0)] - f(x_0))^2 \end{aligned}$$

The second term is the conditional square bias.

(c) Similarly:

$$\mathbb{E}_{\mathcal{Y},\mathcal{X}}[(f(x_0) - \hat{f}(x_0))^2] = \text{Var}_{\mathcal{Y},\mathcal{X}}(\hat{f}(x_0)) + (\mathbb{E}_{\mathcal{Y},\mathcal{X}}[\hat{f}(x_0)] - f(x_0))^2$$

(d) We now use the linearity assumption. For simplicity, we write y and $L = \ell(x_0, \mathcal{X})$ to be the corresponding vectors, so that $\hat{f}(x_0) = L^T y$. Further, we let $f(\mathcal{X}) = (f(x_1), \dots, f(x_n))$ and $\varepsilon = f(\mathcal{X}) - y$. Then

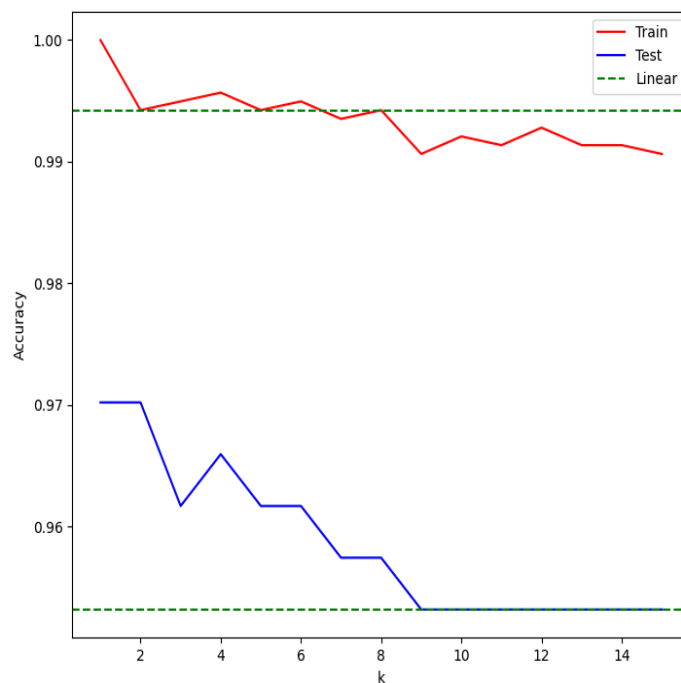
$$\mathbb{E}_{\mathcal{Y}|\mathcal{X}}[\hat{f}(x_0)] - f(x_0) = \mathbb{E}_{\mathcal{Y}|\mathcal{X}}[L^T(f(\mathcal{X}) + \varepsilon)] - f(x_0) = L^T f(\mathcal{X}) - f(x_0)$$

and

$$\text{Var}_{\mathcal{Y}|\mathcal{X}}(\hat{f}(x_0)) = \text{Var}_{\varepsilon}(L^T \varepsilon) = \sigma^2 L^T L$$

Exercise 2.8. Compare the classification performance of linear regression and k nearest neighbor classification on the zipcode data. In particular, consider only the 2s and 3s, and $k = 1, 3, 5, 7$ and 15. Show both the training and test error for each choice. The zipcode data are available from the book website www-stat.stanford.edu/ElemStatLearn.

Solution. The results for different k and linear regression are shown below.



```

import numpy as np
from sklearn.neighbors import KNeighborsClassifier as KNClassifier
from sklearn.linear_model import LinearRegression
from sklearn.metrics import accuracy_score as accuracy
from matplotlib import pyplot as plt

train_data = np.loadtxt('training_data')
train_df = train_data[np.where
    ((train_data[:, 0] == 2) | (train_data[:, 0] == 3))]

test_data = np.loadtxt('test_data')
test_df = test_data[np.where
    ((test_data[:, 0] == 2) | (test_data[:, 0] == 3))]

X_train, X_test = train_df[:, 1:], test_df[:, 1:]
y_train, y_test = train_df[:, 0].reshape(-1), test_df[:, 0].reshape(-1)

k_list = range(1,16)
classifiers = []
for k in k_list:
    classifier = KNClassifier(k, n_jobs = -1)
    classifier.fit(X_train, y_train)
    classifiers.append(classifier)

accs_train = []
accs_test = []
for i in range(len(k_list)):
    y_train_predict = classifiers[i].predict(X_train)
    y_test_predict = classifiers[i].predict(X_test)
    accs_train.append(accuracy(y_train_predict, y_train))
    accs_test.append(accuracy(y_test_predict, y_test))

lin_model = LinearRegression()
lin_model.fit(X_train, y_train)
linear_train_acc = accuracy(y_train, lin_model.predict(X_train).round())
linear_test_acc = accuracy(y_test, lin_model.predict(X_test).round())

```



```

plt.plot(k_list, accs_train, color = 'red', label = 'Train')
plt.plot(k_list, accs_test, color = 'blue', label = 'Test')
plt.axhline(
    linear_train_acc, color = 'green', label = 'Linear', ls = '--')
plt.axhline(linear_test_acc, color = 'green', ls = '--')

plt.xlabel('k')
plt.ylabel('Accuracy')
plt.legend()
plt.show()

```

Exercise 2.9. Consider a linear regression model with p parameters, fit by least squares to a set of training data $(x_1, y_1), \dots, (x_N, y_N)$ drawn at random from a population. Let $\hat{\beta}$ be the least squares estimate. Suppose we have some test data $(\tilde{x}_1, \tilde{y}_1), \dots, (\tilde{x}_M, \tilde{y}_M)$ drawn at random from the same population as the training data. If $R_{tr}(\beta) = \frac{1}{N} \sum_{i=1}^N (y_i - \beta^T x_i)^2$ and $R_{te}(\beta) = \frac{1}{M} \sum_{i=1}^M (\tilde{y}_i - \beta^T \tilde{x}_i)^2$ prove that

$$E[R_{tr}(\hat{\beta})] \leq E[R_{te}(\hat{\beta})]$$

where the expectations are over all that is random in each expression. [This exercise was brought to our attention by Ryan Tibshirani, from a homework assignment given by Andrew Ng.]

Solution. Note that the terms $(\tilde{y}_i - \hat{\beta}^T \tilde{x}_i)^2$ are i.i.d. in $(\tilde{x}_i, \tilde{y}_i)$ since $\hat{\beta}$ only depends on the training data. Thus, the RHS of the inequality is independent of M and we may set $M = N$.

For the test data $(\tilde{x}_1, \tilde{y}_1), \dots, (\tilde{x}_N, \tilde{y}_N)$, let $\tilde{\beta}$ be the least squares estimate of β . By definition $R_{te}(\hat{\beta}) \geq R_{te}(\tilde{\beta})$ so that:

$$E[R_{te}(\hat{\beta})] \geq E[R_{te}(\tilde{\beta})].$$

On the other hand, since the number of samples and the distribution of the test data and training data are the exact same, $R_{te}(\tilde{\beta})$ is equal in distribution to $R_{tr}(\hat{\beta})$. Thus the RHS of the above inequality is in fact equal to $E[R_{tr}(\hat{\beta})]$, and we are done.

3 Linear Methods for Regression

Exercise 3.1. Show that the F statistic (3.13) for dropping a single coefficient from a model is equal to the square of the corresponding z -score (3.12).

Solution. Let $\hat{\mathbf{y}}_0$ and $\hat{\mathbf{y}}_1$ be the estimators of $\mathbf{y}$ with the j -th variable removed and the full model, respectively. Then $\hat{\mathbf{y}}_0$ is the projection of $\mathbf{y}$ onto the span of the $\mathbf{x}_i$ with $i \neq j$ and $\hat{\mathbf{y}}_1$ is the projection of $\mathbf{y}$ onto the span of the $\mathbf{x}_i$. By the Pythagorean Theorem:

$$\text{RSS}_0 = \|\hat{\mathbf{y}}_0 - \mathbf{y}\|^2 = \|\hat{\mathbf{y}}_1 - \mathbf{y}\|^2 + \frac{\langle \mathbf{y}, \mathbf{z}_j \rangle^2}{\|\mathbf{z}_j\|^2} = \text{RSS}_1 + \frac{\langle \mathbf{y}, \mathbf{z}_j \rangle^2}{\|\mathbf{z}_j\|^2},$$

where $\mathbf{z}_j$ is the residual of $\mathbf{x}_j$ regressed on the remaining $\mathbf{x}_i$.

Further, $\text{RSS}_1 = (N - p_1 - 1)\hat{\sigma}^2$ by (3.8) so that:

$$F = (N - p_1 - 1) \cdot \frac{\text{RSS}_0 - \text{RSS}_1}{\text{RSS}_1} = \frac{\langle \mathbf{y}, \mathbf{z}_j \rangle^2}{\hat{\sigma}^2 \|\mathbf{z}_j\|^2} = \frac{\hat{\beta}_j^2 \|\mathbf{z}_j\|^2}{\hat{\sigma}^2},$$

where we used (3.28). To conclude, recall that:

$$\text{Var}(\hat{\beta}_j) = v_j \sigma^2 = \frac{\sigma^2}{\|\mathbf{z}_j\|^2},$$

and use the definition of the z -score.

Exercise 3.2. Given data on two variables X and Y , consider fitting a cubic polynomial regression model $f(X) = \sum_{j=0}^3 \beta_j X^j$. In addition to plotting the fitted curve, you would like a 95% confidence band about the curve. Consider the following two approaches:

1. At each point x_0 , form a 95% confidence interval for the linear function

$$a^T \beta = \sum_{j=0}^3 \beta_j x_0^j$$

2. Form a 95% confidence set for β as in (3.15), which in turn generates confidence intervals for $f(x_0)$.

How do these approaches differ? Which band is likely to be wider? Conduct a small simulation experiment to compare the two methods.

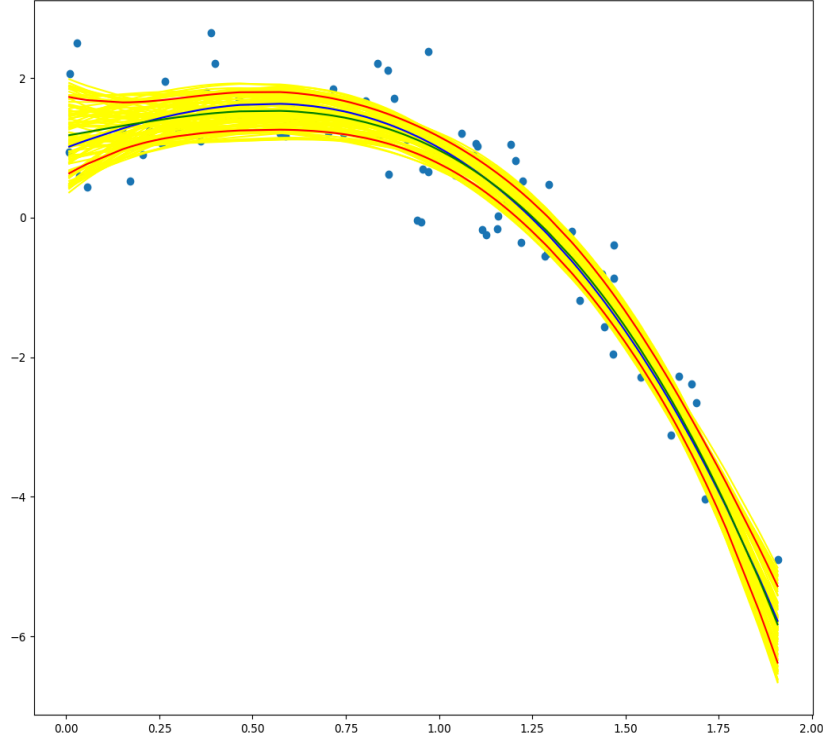


Figure 1: The blue and green lines are the actual and predicted (by OLS) curves for a cubic polynomial. The yellow shaded region are in fact the graphs of 100 β' s obtained by method 2 for different δ uniformly distributed on the sphere of appropriate radius. The red lines are the error bands for method 1.

Solution. For method 1, since $\hat{\beta} \sim N(\beta, (\mathbf{X}^T \mathbf{X})^{-1} \sigma^2)$ then $a^T \hat{\beta} \sim N(a^T \beta, a^T (\mathbf{X}^T \mathbf{X})^{-1} a \sigma^2)$. Hence the 95% confidence interval for $x_0^T \beta$ is:

$$x_0^T \hat{\beta} \pm 1.96 \cdot \sigma \sqrt{x_0^T (\mathbf{X}^T \mathbf{X})^{-1} x_0}.$$

For method 2, the confidence set for β is given by:

$$C_\beta = \{\beta : (\beta - \hat{\beta})^T \mathbf{X}^T \mathbf{X} (\beta - \hat{\beta}) \leq \hat{\sigma}^2 \chi^2(0.95, 4)\}$$

Letting LL^T be the Cholesky decomposition of $\mathbf{X}^T \mathbf{X}$, any β on the boundary of C_β is of the form:

$$\beta = \hat{\beta} + (L^T)^{-1} \delta,$$

where δ is a vector on the sphere of radius $\hat{\sigma} \sqrt{\chi^2(0.95, 4)}$. The error on y is then given by $x_0^T (L^T)^{-1} \delta$. If δ is the most expanded direction (in the SVD) of $(L^T)^{-1}$, this may exceed the error of the former method. We illustrate this in the image above generated from the code that follows.

```

import numpy as np
from matplotlib import pyplot as plt
from scipy.stats import chi2

x_min = 0
x_max = 2
n = 100
sigma = 0.7
p = 0.95
df = 4
num = 100

f = lambda x: 1 + 2 * x - x ** 2 - x ** 3
f_vec = np.vectorize(f)

def make_X(n, x_min, x_max):
    x0 = np.ones(n)
    x1 = np.sort((x_max-x_min) * np.random.random(n) + x_min)
    x2 = np.square(x1)
    x3 = np.power(x1, 3)
    return np.column_stack((x0, x1, x2, x3))

X = make_X(n, x_min, x_max)

y_actual = f_vec(X[:, 1].T)
y_observed = y_actual + sigma*np.random.normal(0, sigma, n)
y_var = np.sqrt(np.diag(X @ np.linalg.inv(X.T @ X) @ X.T))

beta_ols = np.linalg.inv(X.T @ X) @ X.T @ y_observed
y_predict = X @ beta_ols

y_max_1 = y_predict + 1.96 * sigma * y_var
y_min_1 = y_predict - 1.96 * sigma * y_var

L = np.linalg.cholesky(X.T @ X)
U = L.T
U_inv = np.linalg.inv(U)
for i in range(num):
    delta = np.random.normal(0,1,4)
    delta = delta * sigma * np.sqrt(chi2.ppf(p, df))
        / (np.linalg.norm(delta, ord = 2))
    delta = U_inv @ delta
    beta_sim = beta_ols + delta
    plt.plot(X[:, 1], X @ beta_sim, color = 'yellow')

```

```
plt.scatter(X[:, 1], y_observed)
plt.plot(X[:, 1], y_actual, color = 'blue')
plt.plot(X[:, 1], y_predict, color = 'green')
plt.plot(X[:, 1], y_max_1, color = 'red')
plt.plot(X[:, 1], y_min_1, color = 'red')
plt.show()
```

Exercise 3.3. *Gauss-Markov theorem:*

- (a) *Prove the Gauss-Markov theorem: the least squares estimate of a parameter $a^T\beta$ has variance no bigger than that of any other linear unbiased estimate of $a^T\beta$ (Section 3.2.2).*
- (b) *The matrix inequality $B \prec A$ holds if $B - A$ is positive semidefinite. Show that if $\hat{\mathbf{V}}$ is the variance-covariance matrix of the least squares estimate of β and $\tilde{\mathbf{V}}$ is the variance-covariance matrix of any other linear unbiased estimate, then $\hat{\mathbf{V}} \prec \tilde{\mathbf{V}}$.*

Solution.

- (a) Write $c^T = a^T(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T + \Delta^T$. Then:

$$\mathbb{E}(c^T y) = \mathbb{E}((a^T(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T + \Delta^T)y) = a^T\beta + \mathbb{E}(\Delta^T X \beta),$$

and since $c^T y$ is unbiased we have $\Delta^T X = 0$.

For simplicity, let $a^T(\mathbf{X}^T\mathbf{X})^{-1} = M$. Now we compute the variance:

$$\begin{aligned} \text{Var}(c^T y) &= \mathbb{E}(y^T c c^T y) - (a^T \beta)^2 \\ &= \mathbb{E}(y^T (M \mathbf{X}^T + \Delta^T)^T (M \mathbf{X}^T + \Delta^T) y) - (a^T \beta)^2 \\ &= \mathbb{E}(y^T (M \mathbf{X}^T)^T (M \mathbf{X}^T) y) - (a^T \beta)^2 + \mathbb{E}(y^T \Delta \Delta^T y) \\ &= \text{Var}_{OLS} + \mathbb{E}((\Delta^T y)^2), \end{aligned}$$

where on the third line we used $\Delta^T X = 0$. Since $\mathbb{E}((\Delta^T y)^2) \geq 0$, this concludes the proof.

- (b) The proof is the same but in matrix form.

Exercise 3.4. *Show how the vector of least squares coefficients can be obtained from a single pass of the Gram-Schmidt procedure (Algorithm 3.1). Represent your solution in terms of the QR decomposition of $\mathbf{X}$.*

Solution. By (3.32), $\hat{\beta} = \mathbf{R}^{-1}\mathbf{Q}^T\mathbf{y}$. Since $\mathbf{Q} = \mathbf{ZD}^{-1}$, $\mathbf{Q}$ can be computed as one obtains $\mathbf{z}_i$ in the Gram-Schmidt algorithm by entering the normalized $\mathbf{z}_i$. Similarly, $\mathbf{R} = \mathbf{D}\mathbf{\Gamma}$, is immediately found as the entries of $\hat{\gamma}_{kj}$ with the normalized $\mathbf{z}_i$. Since $\mathbf{R}$ is upper triangular, it can be inverted quickly.

Exercise 3.5. Consider the ridge regression problem (3.41). Show that this problem is equivalent to the problem

$$\hat{\beta}^c = \arg \min_{\beta^c} \left\{ \sum_{i=1}^N [y_i - \beta_0^c - \sum_{j=1}^p (x_{ij} - \bar{x}_j) \beta_j^c]^2 + \lambda \sum_{j=1}^p \beta_j^{c2} \right\}$$

Give the correspondence between β^c and the original β in (3.41). Characterize the solution to this modified criterion. Show that a similar result holds for the lasso.

Solution. Letting $\beta_0 = \beta_0^c - \sum_{j=1}^p \bar{x}_j \beta_j^c$, and $\beta_i = \beta_i^c$ for $i \neq 0$, the problem above becomes the usual lasso

$$\hat{\beta} = \arg \min_{\beta} \left\{ \sum_{i=1}^N [y_i - \beta_0 - \sum_{j=1}^p x_{ij} \beta_j]^2 + \lambda \sum_{j=1}^p \beta_j^2 \right\}$$

From the relations between β and β^c , we see that the slope coefficients for the centered problem are the same, whereas the intercept $\beta_0^c = \beta_0 + \bar{\mathbf{x}}^T \beta$ is the predicted value (by the usual lasso) at the mean of the data. The analysis for ridge is identical, since it only changes the penalty term.

Exercise 3.6. Show that the ridge regression estimate is the mean (and mode) of the posterior distribution, under a Gaussian prior $\beta \sim N(0, \tau^2 \mathbf{I})$, and Gaussian sampling model $\mathbf{y} \sim N(\mathbf{X}\beta, \sigma^2 \mathbf{I})$. Find the relationship between the regularization parameter λ in the ridge formula, and the variances τ and σ^2 .

Solution. By Bayes' formula:

$$f(\beta|\mathbf{y}, \mathbf{X}) \propto f(\mathbf{y}, \mathbf{X}|\beta) f(\beta) \propto f(\mathbf{y}|\mathbf{X}, \beta) f(\beta),$$

since $f(\mathbf{X})$ does not depend on β . Thus:

$$\begin{aligned} f(\beta|\mathbf{y}, \mathbf{X}) &\propto \exp \left(-\frac{(\mathbf{y} - \mathbf{X}\beta)^T (\mathbf{y} - \mathbf{X}\beta)}{2\sigma^2} - \frac{1}{2\tau^2} \beta^T \beta \right) \\ &\propto \exp \left(\frac{-1}{2\sigma^2} \left(-\beta^T \left(\mathbf{X}^T \mathbf{X} + \frac{\sigma^2}{\tau^2} \mathbf{I} \right) \beta + \mathbf{y}^T \mathbf{X} \beta + \beta^T \mathbf{X}^T \mathbf{y} \right) \right) \end{aligned}$$

Recall in general that a multivariate Gaussian in β with mean μ and covariance matrix Σ has pdf proportional to

$$\exp \left(-\frac{(\beta - \mu)^T \Sigma^{-1} (\beta - \mu)}{2} \right) \propto \exp \left(-\frac{\beta^T \Sigma^{-1} \beta + \beta^T \Sigma^{-1} \mu + \mu^T (\Sigma^{-1})^T \beta}{2} \right)$$

Comparing the two, we see that the pdf $f(\beta|\mathbf{y}, \mathbf{X})$ is indeed Gaussian with

$$\Sigma^{-1} = \frac{1}{\sigma^2} \left(\mathbf{X}^T \mathbf{X} + \frac{\sigma^2}{\tau^2} \mathbf{I} \right),$$

and moreover we see that $\mathbf{X}^T \mathbf{y} = \sigma^2 \Sigma^{-1} \mu$ by comparing the third term in $f(\beta|\mathbf{y}, \mathbf{X})$ with the second term in the general Gaussian pdf. Solving for μ and plugging Σ^{-1} found as above gives:

$$\mu = \left(\mathbf{X}^T \mathbf{X} + \frac{\sigma^2}{\tau^2} \mathbf{I} \right)^{-1} \mathbf{X}^T \mathbf{y},$$

which is precisely β^{ridge} with $\lambda = \sigma^2/\tau^2$, as desired.

Exercise 3.7. Consider the decomposition of the uncentered $N \times (p+1)$ matrix $\mathbf{X}$ whose first column is all ones), and the SVD of the $N \times p$ centered matrix $\tilde{\mathbf{X}}$. Show that $\mathbf{Q}_2$ and $\mathbf{U}$ span the same subspace, where $\mathbf{Q}_2$ is the sub-matrix of $\mathbf{Q}$ with the first column removed. Under what circumstances will they be the same, up to sign flips?

Solution. Recall that $\mathbf{Q} = \mathbf{ZD}^{-1}$, where $\mathbf{Z}$ is the vector of residuals obtained in the Gram-Schmidt algorithm. Hence, $\mathbf{q}_0$ is a scalar multiple of $\mathbf{1}$, since $\mathbf{z}_0 = \mathbf{1}$. Since $\mathbf{Q}$ is orthogonal, the span of $\mathbf{Q}_2$ is equal to $\text{span}(\mathbf{Q}) \cap \mathbf{1}^\perp$, where $\perp$ denotes orthogonal complement. Moreover $\text{span}(\mathbf{Q}) = \text{span}(\mathbf{X})$, since $\mathbf{R}$ is invertible. Hence $\text{span}(\mathbf{Q}_2) = \text{span}(\mathbf{X}) \cap \mathbf{1}^\perp$.

Clearly also $\text{span}(\mathbf{U}) = \text{span}(\tilde{\mathbf{X}})$, since again $\mathbf{DV}$ is invertible. But note that $\tilde{\mathbf{x}}_i$ is simply the projection of $\mathbf{x}_i$ onto $\mathbf{1}^\perp$, since $\tilde{\mathbf{x}}_i = \mathbf{x}_i - \bar{x}_i \mathbf{1}$, where $\bar{x}_i = (1/N) \mathbf{x}_i^T \mathbf{1}$. Hence $\text{span}(\mathbf{U}) = \text{span}(\tilde{\mathbf{X}}) = \text{span}(\mathbf{X}) \cap \mathbf{1}^\perp$ and the first part of the exercise is complete.

For these to agree, we need $\mathbf{DV} = \mathbf{R}$, i.e., an orthogonal matrix is equal to an upper-triangular one. Thus both have to be the identity (up to sign flips) and $\mathbf{X} = \mathbf{Q}$ (up to sign flips), i.e. the feature input vectors $\mathbf{x}_i$ must have been orthogonal to begin with.

Exercise 3.8. Forward stepwise regression. Suppose we have the QR decomposition for the $N \times p$ matrix $\mathbf{X}_1$ in a multiple regression problem with response $\mathbf{y}$, and we have an additional $p-q$ predictors in the matrix $\mathbf{X}_2$. Denote the current residual by $\mathbf{r}$. We wish to establish which one of these additional variables will reduce the residual sum-of-squares the most when included with those in $\mathbf{X}_1$. Describe an efficient procedure for doing this.

Solution. Let $\mathbf{y}_0$ be the current predictor of $\mathbf{y}$, i.e., $\mathbf{y} = \mathbf{y}_0 + \mathbf{r}$. Then $\mathbf{y}_0$ is the orthogonal projection of $\mathbf{y}$ onto the column span of $\mathbf{Q}$. If we add a new column $\mathbf{x}_{p+1}$, the new residual will be given by the projection of $\mathbf{R}$ we on the span columns of $\mathbf{Q}$ and $\mathbf{x}$. Letting $\mathbf{z}_{p+1} = \mathbf{x}_{p+1} - \sum_{i=1}^p \langle \mathbf{x}_{p+1}, \mathbf{q}_i \rangle \mathbf{q}_i$ be the orthogonal component of $\mathbf{x}_{p+1}$ relative to the column span of $\mathbf{Q}$, we can write the residue $\mathbf{r}$ as

$$\mathbf{r} = \mathbf{r}' + \frac{\langle \mathbf{r}, \mathbf{z}_{p+1} \rangle}{\langle \mathbf{z}_{p+1}, \mathbf{z}_{p+1} \rangle} \mathbf{z}_{p+1},$$

where $\mathbf{r}'$ will be the new residues after adding $\mathbf{x}_{p+1}$. Hence the difference in RSS is given by $|\langle \mathbf{r}, \mathbf{z}_{p+1} \rangle| / \|\mathbf{z}_{p+1}\|$. We can perform this procedure for all columns of $\mathbf{X}_2$ and find the largest value.

Note that this also allows us to update the QR decomposition for the next step as well via $\mathbf{q}_{p+1} = \mathbf{z}_{p+1} / \|\mathbf{z}_{p+1}\|$.

Exercise 3.9. *Backward stepwise regression. Suppose we have the multiple regression fit of $\mathbf{X}$ on $\mathbf{y}$, along with the standard errors and Z -scores as in Table 3.2. We wish to establish which variable, when dropped, will increase residual sum-of-squares the least. How would you do this?*

Solution. By Exercise 3.1, the residual sum of squares difference (F -score) is proportional to the Z -score when dropping a single variable. Hence we simply pick the variable with smallest Z -score to drop.

Exercise 3.10. *Show that the ridge regression estimates can be obtained by ordinary least squares regression on an augmented data set. We augment the centered matrix $\mathbf{X}$ with p additional rows $\sqrt{\lambda}\mathbf{I}$, and augment $\mathbf{y}$ with p zeros. By introducing artificial data having response value zero, the fitting procedure is forced to shrink the coefficients toward zero. This is related to the idea of hints due to Abu-Mostafa (1995), where model constraints are implemented by adding artificial data examples that satisfy them.*

Solution. The ordinary least squares regression solves:

$$\hat{\beta}^{OLS} = \arg \min_{\beta} \|\mathbf{y} - \mathbf{X}\beta\|^2$$

For the augmented matrix $\mathbf{X}$ and the augmented $\mathbf{y}$, by the Pythagorean theorem this is simply:

$$\hat{\beta}^{ridge} = \arg \min_{\beta} \|\mathbf{y} - \mathbf{X}\beta\|^2 + \|\mathbf{0} - \sqrt{\lambda}\beta\|^2 = \arg \min_{\beta} \|\mathbf{y} - \mathbf{X}\beta\|^2 + \lambda\|\beta\|^2,$$

which is the Ridge regression estimator.

Exercise 3.11. Consider a regression problem with all variables and response having mean zero and standard deviation one. Suppose also that each variable has identical absolute correlation with the response:

$$\frac{1}{N} |\langle \mathbf{x}_j, \mathbf{y} \rangle| = \lambda, \quad j = 1, \dots, p.$$

Let $\hat{\beta}$ be the least-squares coefficient of $\mathbf{y}$ on $\mathbf{X}$ and let $\mathbf{u}(\alpha) = \alpha \mathbf{X} \hat{\beta}$ for $\alpha \in [0, 1]$ be the vector that moves a fraction α toward the least squares fit $\mathbf{u}$. Let RSS be the residual sum-of-squares from the full least squares fit.

(a) Show that

$$\frac{1}{N} |\langle \mathbf{x}_j, \mathbf{y} - \mathbf{u}(\alpha) \rangle| = (1 - \alpha) \lambda, \quad j = 1, \dots, p,$$

and hence the correlations of each $\mathbf{x}_j$ with the residuals remain equal in magnitude as we progress toward $\mathbf{u}$.

(b) Show that these correlations are all equal to

$$\lambda(\alpha) = \frac{(1 - \alpha)}{\sqrt{(1 - \alpha)^2 + \frac{\alpha(2 - \alpha)}{N} \cdot RSS}} \lambda,$$

and hence they decrease monotonically to zero.

(c) Use these results to show that the LAR algorithm in Section 3.4.4 keeps the correlations tied and monotonically decreasing, as claimed in (3.55)

Solution.

- (a) Write $\mathbf{u}(\alpha) = \mathbf{u} - (1 - \alpha) \mathbf{X} \hat{\beta}$, and recall that $\langle \mathbf{x}_j, \mathbf{y} - \mathbf{u} \rangle = 0$ since the residuals are orthogonal to the predictors for the OLS solution, so that

$$|\langle \mathbf{x}_j, \mathbf{y} - \mathbf{u}(\alpha) \rangle| = |\langle \mathbf{x}_j, (1 - \alpha) \mathbf{X} \hat{\beta} \rangle| = (1 - \alpha) |\langle \mathbf{x}_j, \mathbf{X} \hat{\beta} \rangle|.$$

On the other hand, $\langle \mathbf{x}_j, \mathbf{X} \hat{\beta} \rangle = \langle \mathbf{x}_j, \mathbf{X} \hat{\beta} \rangle + \langle \mathbf{x}_j, \mathbf{r} \rangle = \langle \mathbf{x}_j, \mathbf{X} \hat{\beta} + \mathbf{r} \rangle = \langle \mathbf{x}_j, \mathbf{y} \rangle$, since once again the residual $\mathbf{r} = \mathbf{y} - \mathbf{u}$ is orthogonal to $\mathbf{x}_j$. Using the hypothesis concludes the proof.

- (b) The correlation is given by

$$N \frac{(1 - \alpha) \lambda}{\|\mathbf{x}_j\| \|\mathbf{y} - \mathbf{u}(\alpha)\|},$$

using part (a). Since $\|\mathbf{x}_j\|^2 = N$ by hypothesis, we only have to compute $\|\mathbf{y} - \mathbf{u}(\alpha)\|$. Recall that $\|\mathbf{y}\|^2 = N$, since the response is also assumed to be standardized, and since $\mathbf{u}$ and $\mathbf{r}$ are orthogonal, we see that $\|\mathbf{u}\|^2 = N - RSS$, since $RSS = \|\mathbf{r}\|^2$ by definition.

Again, write $\mathbf{y} = \mathbf{u} + \mathbf{r}$, so that $\mathbf{y} - \mathbf{u}(\alpha) = (1 - \alpha)\mathbf{u} + \mathbf{r}$ and so again by orthogonality:

$$\begin{aligned}\|\mathbf{y} - \mathbf{u}(\alpha)\|^2 &= (1 - \alpha)^2\|\mathbf{u}\|^2 + \|\mathbf{r}\|^2 \\ &= (1 - \alpha)^2(N - \text{RSS}) + \text{RSS} \\ &= (1 - \alpha)^2N + \alpha(2 - \alpha)\text{RSS}.\end{aligned}$$

Plugging and canceling the N 's out gives the desired formula. Clearly as $\alpha \rightarrow 1$ the numerator goes to 0 whereas the denominator does not, due to the second term.

- (c) Suppose inductively that at the beginning of the current step the active variables $\mathcal{A}_k$ have the same correlation λ with the residual $\mathbf{r}_k = \mathbf{y} - \mathbf{X}_{\mathcal{A}_k}\beta_{\mathcal{A}_k}$. Since $\beta_{\mathcal{A}_k}(\alpha) = \beta_{\mathcal{A}_k} + \alpha\delta_k$, and δ_k is obtained as the regression coefficient vector of $\mathbf{r}_k$ on the covariates in $\mathcal{A}_k$, the new residual as a function of α is:

$$\mathbf{r}_k(\alpha) = \mathbf{y} - \mathbf{X}_{\mathcal{A}_k}(\beta_{\mathcal{A}_k} + \alpha\delta_k) = \mathbf{r}_k - \alpha\mathbf{X}_{\mathcal{A}_k}\delta_k.$$

By (a), the correlations of $\mathbf{r}_k(\alpha)$ with the covariates in $\mathcal{A}_k$ remain the same.

Exercise 3.12. *LAR directions. Using the notation around equation (3.55) on page 74, show that the LAR direction makes an equal angle with each of the predictors in $\mathcal{A}_k$.*

Solution. Since the LAR direction $\mathbf{u}$ is obtained by projecting $\mathbf{r}$ onto the span of the variables in $\mathcal{A}_k$, evidently $\langle \mathbf{x}_j, \mathbf{u} \rangle = \langle \mathbf{x}_j, \mathbf{r} \rangle$, for $\mathbf{x}_j \in \mathcal{A}_k$. By the previous exercise, at each step of the LAR the residual $\mathbf{r}$ makes the same angle with the variables $\mathbf{x}_j \in \mathcal{A}_k$, since they are normalized to unit, hence so does $\mathbf{u}$.

Exercise 3.13. *LAR look-ahead (Efron et al., 2004, Sec. 2 Least Angle Regression). Starting at the beginning of the k -th step of the LAR algorithm, derive expressions to identify the next variable to enter the active set at step $k + 1$, and the value of α at which this occurs (using the notation around equation (3.55) on page 74).*

Solution. Again, denote the current residue by $\mathbf{r}$, the LAR direction

$$\mathbf{u} = \mathbf{X}_{\mathcal{A}_k}(\mathbf{X}_{\mathcal{A}_k}^T \mathbf{X}_{\mathcal{A}_k})^{-1} \mathbf{X}_{\mathcal{A}_k}^T \mathbf{r}$$

and the active variables by $\mathcal{A}_k$. Then, as before:

$$\frac{1}{N} |\langle \mathbf{x}_j, \mathbf{r} - \alpha \mathbf{u} \rangle| = (1 - \alpha)\lambda, \quad \text{for } \mathbf{x}_j \in \mathcal{A}_k,$$

where $\lambda = |\langle \mathbf{x}_j, \mathbf{r} \rangle|$. For $\mathbf{x}_i \notin \mathcal{A}_k$, we have $|\langle \mathbf{x}_i, \mathbf{r} - \alpha \mathbf{u} \rangle| = |\langle \mathbf{x}_i, \mathbf{r} \rangle - \alpha \langle \mathbf{x}_i, \mathbf{u} \rangle|$, where the sign depends on whether the two inner products on the right hand side have the same sign.

Equating the two expressions, gives

$$\alpha_i = \frac{\langle \mathbf{x}_i, \mathbf{r} \rangle - \lambda}{\langle \mathbf{x}_i, \mathbf{u} \rangle - \lambda} \text{ or } \alpha_i = \frac{\langle \mathbf{x}_i, \mathbf{r} \rangle + \lambda}{\langle \mathbf{x}_i, \mathbf{u} \rangle + \lambda},$$

depending on which of the expressions is positive. Finding the minimal positive α_i then yields $\mathbf{x}_i$ which should enter in the next step.

Exercise 3.14.

Exercise 3.15 (Lasso and LAR). Consider the lasso problem in Lagrange multiplier form: with $L(\beta) = \frac{1}{2} \sum_i (y_i - \sum_j x_{ij} \beta_j)^2$, we minimize

$$L(\beta) + \sum_j \lambda |\beta_j| \quad (1)$$

for fixed $\lambda > 0$.

- (a) Setting $\beta_j = \beta_j^+ - \beta_j^-$, with $\beta_j^+, \beta_j^- \geq 0$, expression (1) becomes $L(\beta) + \lambda \sum_j (\beta_j^+ - \beta_j^-)$. Show that the Lagrange dual function is

$$L(\beta) + \lambda \sum_j (\beta_j^+ - \beta_j^-) - \sum_j \lambda_j^+ \beta_j^+ - \sum_j \lambda_j^- \beta_j^-,$$

and the Karush-Kuhn-Tucker optimality conditions are:

$$\begin{aligned} \nabla L(\beta)_j + \lambda - \lambda_j^+ &= 0 \\ -\nabla L(\beta)_j + \lambda - \lambda_j^- &= 0 \\ \lambda_j^+ \beta_j^+ &= 0 \\ \lambda_j^- \beta_j^- &= 0 \end{aligned}$$

along with non-negativity constraints on the parameters and all the Lagrange multipliers.

- (b) Show that $|\nabla L(\beta)_j| \leq \lambda$ for all j , and that the KKT conditions imply one of the following three scenarios:

$$\begin{aligned} \lambda = 0 &\implies \nabla L(\beta)_j = 0, \forall j \\ \beta_j^+ > 0, \lambda > 0 &\implies \lambda_j^+ = 0, \nabla L(\beta)_j = -\lambda < 0, \beta_j^- = 0 \\ \beta_j^- > 0, \lambda > 0 &\implies \lambda_j^- = 0, \nabla L(\beta)_j = \lambda > 0, \beta_j^+ = 0 \end{aligned}$$

- (c) Suppose the set of active predictors is unchanged for $\lambda_0 \geq \lambda \geq \lambda_1$. Show that there is a vector γ_0 such that

$$\hat{\beta}(\lambda) = \hat{\beta}(\lambda_0) - (\lambda - \lambda_0)\gamma_0.$$

Thus the lasso solution path is linear as λ ranges from λ_0 to λ_1 (Efron et al., 2004; Rosset and Zhu, 2007).

Solution.

- (a) We have $2p$ variables $\beta_j^\pm$ and constraints $-\beta_j^\pm \leq 0$. Recall that the Lagrangian is obtained by adding each of the constraints to the objective function multiplied by a multiplier, which gives the desired expression. The first two KKT conditions stated are the stationarity conditions, obtained by taking gradients with respect to $\beta_j^\pm$. The

remaining two are known as complementary slackness conditions.

- (b) The condition $|\nabla L(\beta)_j| \leq \lambda$ follows immediately from the three scenarios that follows plus the observation that if $\beta_j^+ = \beta_j^- = 0$ then $\nabla L(\beta)_j = 0$.

If $\lambda = 0$, that is scenario 1, for all j we have $\nabla L(\beta)_j = \lambda_j^+ = -\lambda_j^-$. Since $\lambda_j^+, \lambda_j^- \geq 0$, this forces $\nabla L(\beta)_j = 0$.

If $\lambda > 0$ and $\beta_j^+ > 0$, that is scenario 2, complementary slackness forces $\lambda_j^+ = 0$, so that the first stationarity condition gives $\nabla L(\beta)_j = -\lambda$. The second stationarity condition then gives $\lambda_j^- = 2\lambda > 0$, so that by complementary slackness $\beta_j^- = 0$. The third scenario is analogous.

- (c) Since the objective function is convex (in fact, strictly convex as long as no variables are only zeros), it suffices to show that there exists a solution to the KKT conditions of the form $\hat{\beta}(\lambda) = \hat{\beta}(\lambda_0) - (\lambda - \lambda_0)\gamma_0$.

Let us analyze the j -th coefficient, and show that it is affine in λ . Without loss of generality, assume that at $\lambda = \lambda_0$ we have $\lambda_j^- = 0$, so that $\nabla L(\beta(\lambda_0))_j = \lambda_0$. Recall that $\nabla L(\beta)_j = \mathbf{x}_j^T(\mathbf{y} - \mathbf{X}\beta)$, so that for a KKT solution we must have

$$\nabla L(\hat{\beta}(\lambda))_j = \mathbf{x}_j^T(\mathbf{y} - \mathbf{X}\hat{\beta}(\lambda)) = \lambda$$

for $\lambda_0 \geq \lambda \geq \lambda_1$. Choosing the appropriate $\hat{\beta}(\lambda_0)$ so that the expression holds at λ_0 , and plugging $\hat{\beta}(\lambda) = \hat{\beta}(\lambda_0) - (\lambda - \lambda_0)\gamma_0$, gives

$$(\lambda - \lambda_0)\mathbf{x}_j^T \mathbf{X}_r \gamma_0 = \lambda - \lambda_0.$$

This in turn gives the solution $\gamma_0 = (\mathbf{X}_r^T \mathbf{X}_r)^{-1}(1, \dots, 1)^T$, where $\mathbf{X}_r$ is the feature matrix of only the active variables.

Exercise 3.16. Suppose for a given t in (3.51) the fitted lasso coefficient for the variable X_j is $\hat{\beta}_j = a$. Suppose we augment our set of variables with an identical copy $X_j^* = X_j$. Characterize the effect of this exact collinearity by describing the set of solutions for $\hat{\beta}_j$ and $\hat{\beta}_j^*$ using the same value of t .

Solution. We will consider adding a copy of the last column $\mathbf{x}_p$. The constrained form of lasso regression is

$$\hat{\beta}^{lasso} = \arg \min_{\beta} \sum_{i=1}^N \left(y_i - \beta_0 - \sum_{j=1}^p x_{ij} \beta_j \right)^2 = \arg \min_{\beta} \sum_{i=1}^N \left(y_i - \beta_0 - \sum_{j=1}^{p-1} x_{ij} \beta_j - x_{ip} \beta_p \right)^2.$$

subject to $\sum_{j=1}^p |\beta_j| \leq t$. By augmenting with a copy as explained, we add a column $x_{i(p+1)} = x_{ip}$ with coefficient β_{p+1} , so the new optimization problem is:

$$\hat{\beta}^{lasso} = \arg \min_{\beta} \sum_{i=1}^N \left(y_i - \beta_0 - \sum_{j=1}^{p-1} x_{ij} \beta_j - x_{ip}(\beta_p + \beta_{p+1}) \right)^2$$

subject to $\sum_{j=1}^{p+1} |\beta_j| \leq t$. Let β_p^0 be the estimator for β_p for the original problem without the copied variable. For the new problem, as long as $\beta_p + \beta_{p+1} = \beta_p^0$ and $|\beta_p| + |\beta_{p+1}| = |\beta_p^0|$ with other coefficients being equal then we have a solution to the new optimization problem which is on the boundary.

Therefore adding the copy of a variable creates ambiguity on the coefficients of the copied variable and its copy, i.e., any choice of the form $\beta_p = t\beta_p^0$, $\beta_{p+1} = (1-t)\beta_p^0$ for $t \in [0, 1]$ will be a solution to lasso regression.

Exercise 3.17. Suppose we run a ridge regression with parameter λ on a single variable X , and get coefficient a . We now include an exact copy X^* , and refit our ridge regression. Show that both coefficients are identical, and derive their value. Show in general that if m copies of a variable X are included in a ridge regression, their coefficients are all the same.

Solution. For the univariate case with one copy we can use the explicit solution $\beta = (\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^T \mathbf{y}$. Note that without the copy we have

$$a = \frac{\langle X, \mathbf{y} \rangle}{\langle X, X \rangle + \lambda}.$$

With the copy, $\mathbf{X}^T \mathbf{X}$ is a 2×2 matrix with $\langle X, X \rangle$ in all 4 entries. Then we can compute directly:

$$(\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I})^{-1} = \frac{1}{2\lambda \langle X, X \rangle + \lambda^2} \begin{pmatrix} \langle X, X \rangle + \lambda & -\langle X, X \rangle \\ -\langle X, X \rangle & \langle X, X \rangle + \lambda \end{pmatrix}$$

Since $\mathbf{X}^T \mathbf{y}$ is a vector with two identical components $\langle X, \mathbf{y} \rangle$, we deduce that the coefficients in β are identical and equal to:

$$\frac{\langle X, \mathbf{y} \rangle}{2\langle X, X \rangle + \lambda}.$$

We use the same notation as in the previous exercise and deal with the general case. We add columns $x_{i(p+1)} = \dots = x_{i(p+m)} = x_{ip}$ with coefficients $\beta_{p+1}, \dots, \beta_{p+m}$, which are copies of the last column $\mathbf{x}_p$. The optimization problem is then:

$$\hat{\beta}^{ridge} = \arg \min_{\beta} \sum_{i=1}^N \left(y_i - \beta_0 - \sum_{j=1}^{p-1} x_{ij} \beta_j - x_{ip} (\beta_p + \dots + \beta_{p+m}) \right)^2$$

subject to $\sum_{j=1}^{p+m} \beta_j^2 \leq t$. Observe that the objective function only depends on the sum $\beta_p + \dots + \beta_{p+m}$. We claim that these $m+1$ coefficients have to be equal. Indeed, suppose that at the optimal solution, where $\sum_{j=1}^{p+m} \beta_j^2 = t$, they are not all equal. The maximum of $x_1 + \dots + x_{m+1}$ subject to $x_1^2 + \dots + x_m^2 \leq t$ is attained when all values are equal, as is easily seen with Lagrange's method. This in turn implies that the solution is not optimal, since we could obtain the same mean squared error while decreasing $\sum_{j=1}^{p+m} \beta_j^2$.

Exercise 3.18. Consider the elastic-net optimization problem:

$$\min_{\beta} \|\mathbf{y} - \mathbf{X}\beta\|^2 + \lambda[\alpha\|\beta\|_2^2 + (1 - \alpha)\|\beta\|_1]$$

Show how one can turn this into a lasso problem, using an augmented version of $\mathbf{X}$ and $\mathbf{y}$.

Solution. We augment $\mathbf{X}$ into $\mathbf{X}'$, a $(N + p) \times p$ matrix, by stacking $\sqrt{\lambda\alpha}\mathbf{I}_p$, where $\mathbf{I}_p$ is the $(p \times p)$ -identity matrix, below $\mathbf{X}$. We augment $\mathbf{y}$ into $\mathbf{y}'$ by adding p zeros at the end. Then $\mathbf{y}' - \mathbf{X}'\beta = (\mathbf{y} - \mathbf{X}\beta, \sqrt{\lambda\alpha}\beta)^T$, so that $\|\mathbf{y}' - \mathbf{X}'\beta\|_2^2 = \|\mathbf{y} - \mathbf{X}\beta\|^2 + \lambda\alpha\|\beta\|_2^2$. Thus, for the augmented problem, elastic-net corresponds to lasso with regularization parameter $\lambda(1 - \alpha)$.

7 Model Assessment and Selection

Exercise 7.1. *Derive the estimate of in-sample error (7.24).*

Solution. The estimate (7.24) is obtained by plugging (7.23) into (7.22). It remains to derive (7.23), assuming additive errors $Y = f(X) + \varepsilon$ and a linear model $\hat{Y} = \mathbf{S}Y$. In what follows since we deal with in-sample errors all expectations are taken over Y , i.e., X is assumed to be fixed. Note that

$$\sum_{i=1}^N \text{Cov}(\hat{y}_i, y_i) = \text{trace}(\text{Cov}(\hat{Y}, Y)),$$

so that it remains to expand the covariance matrix:

$$\begin{aligned} \text{trace}(\text{Cov}(\hat{Y}, Y)) &= \text{trace}(\mathbb{E}[(\hat{Y} - \mathbb{E}(\hat{Y}))(Y - \mathbb{E}[Y])^T]) \\ &= \text{trace}(\mathbb{E}[(\mathbf{S}(f(X) + \varepsilon) - \mathbb{E}(\mathbf{S}(f(X) + \varepsilon)))(f(X) + \varepsilon - \mathbb{E}[f(X) + \varepsilon])^T]) \\ &= \text{trace}(\mathbb{E}[(\mathbf{S}(f(X) + \varepsilon) - \mathbb{E}[\mathbf{S}f(X)])(f(X) + \varepsilon - \mathbb{E}[f(X)])^T]) \\ &= \text{trace}(\mathbb{E}[(\mathbf{S}(f(X) + \varepsilon) - \mathbb{E}[\mathbf{S}f(X)])\varepsilon^T]) \\ &= \text{trace}(\mathbb{E}[\mathbf{S}\varepsilon\varepsilon^T]), \end{aligned}$$

since all the terms not involving ε cancel out and $\mathbb{E}[\varepsilon] = 0$. Using the linearity of expectation:

$$\text{trace}(\mathbb{E}[\mathbf{S}\varepsilon\varepsilon^T]) = \text{trace}(\mathbf{S}\mathbb{E}[\varepsilon\varepsilon^T]) = \sigma_\varepsilon^2 \text{trace}(\mathbf{S}),$$

since ε is assumed to have mean 0 and variance $\sigma_\varepsilon^2 I_N$. By definition $d = \text{trace}(\mathbf{S})$, which completes the proof.

Exercise 7.2. Consider the in-sample prediction error (7.18) and the training error $\overline{err}$ in the case of squared-error loss:

$$Err_{in} = \frac{1}{N} \sum_{i=1}^N E_{Y^0} (Y_i^0 - \hat{f}(x_i))^2$$

$$\overline{err} = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{f}(x_i))^2$$

Add and subtract $f(x_i)$ and $E\hat{f}(x_i)$ in each expression and expand. Hence establish that the average optimism in the training error is

$$\frac{2}{N} \sum_{i=1}^N Cov(\hat{y}_i, y_i),$$

as given in (7.21).

Solution. Write $\hat{y}_i = \hat{f}(x_i)$. Adding and subtracting we can group:

$$Y_i^0 - \hat{f}(x_i) = (Y_i^0 - f(x_i)) + (f(x_i) - E\hat{y}_i) + (E\hat{y}_i - \hat{y}_i)$$

$$y_i - \hat{f}(x_i) = (y_i - f(x_i)) + (f(x_i) - E\hat{y}_i) + (E\hat{y}_i - \hat{y}_i)$$

Squaring these terms and subtracting and taking E_{Y^0} of the first gives:

$$E_{Y^0} (Y_i^0 - \hat{f}(x_i))^2 - (y_i - \hat{f}(x_i))^2 = E_{Y^0} (Y_i^0 - f(x_i))^2 - (y_i - f(x_i))^2 - 2(y_i - f(x_i))[(f(x_i) - E\hat{y}_i) + (E\hat{y}_i - \hat{y}_i)],$$

since $E_{Y^0} (Y_i^0 - \hat{f}(x_i)) = 0$ and the terms involving only the second and third elements of the sum cancel out. Hence the i -th term of the sum $\omega = E_{\mathbf{y}}(Err_{in} - \overline{err})$ is

$$E_{Y^0} (Y_i^0 - f(x_i))^2 - E_{\mathbf{y}} (y_i - f(x_i))^2 + 2E_{\mathbf{y}} [(y_i - f(x_i))[(f(x_i) - E\hat{y}_i) + (\hat{y}_i - E\hat{y}_i)]]$$

The first and second term cancel since Y^0 is sampled from the same distribution as $\mathbf{y}$ and note that

$$2E_{\mathbf{y}} [(y_i - f(x_i))(f(x_i) - E\hat{y}_i)] = 0,$$

since the second term in the product is constant and $E_{\mathbf{y}} (y_i - f(x_i)) = 0$. Hence what is left is:

$$2E_{\mathbf{y}} [(y_i - f(x_i))(\hat{y}_i - E\hat{y}_i)] = Cov(y_i, \hat{y}_i).$$

Adding these up finishes the proof.